Examiner only

8. The word equation for aerobic respiration is shown below.

glucose + oxygen ⟶ carbon dioxide + water + ATP

(a) In the space below state the word equation for anaerobic respiration in human cells. [1]

....

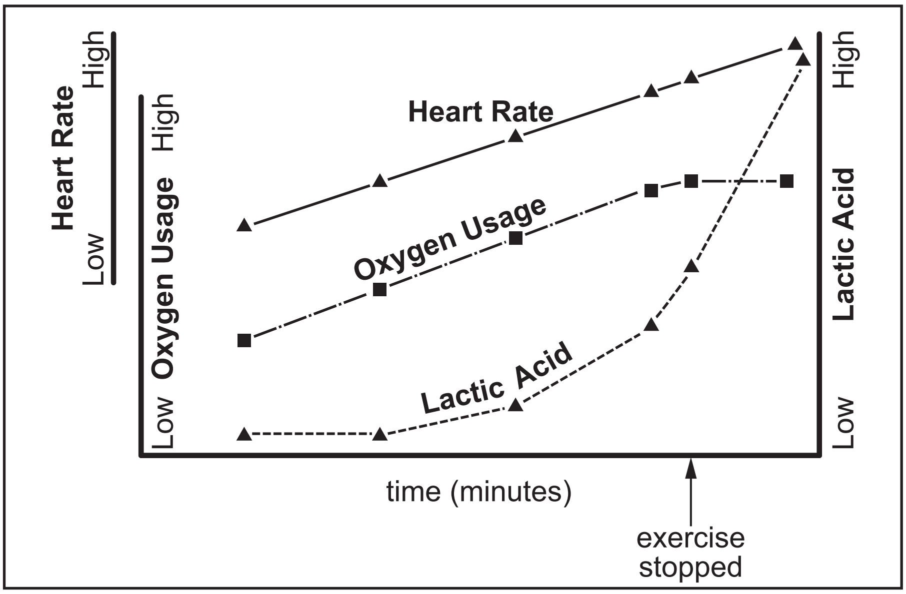
(b) Explain why anaerobic respiration is less efficient than aerobic respiration. [2]

....

....

....

(c) An Olympic athlete exercised on a treadmill. During the exercise her blood oxygen and lactic acid levels were continuously monitored as was her heart rate.
The athlete's fitness coach knew the maximum intensity of exercise she could perform (100%). The athlete increased the intensity of exercise until she reached the maximum intensity of exercise she could perform (100%). She then stopped the exercise but her heart rate and blood levels continued to be monitored.
The graph below appeared on a computer screen.



 (3430UA0-1)

Examiner only

(i) Explain why, even after exercise stops, the athlete continues to take in and use large volumes of oxygen. [2]

....

....

....

(ii) Explain why the heart rate remains high after exercise stops. [1]

....

....

....

(d) The ability of oxygen to pass into the capillaries around the alveoli decreases with increasing altitude. Above 1500 m physical activity becomes more difficult.
In 1968, the Olympic Games were held in Mexico City (altitude 2268 m). Athletes and their coaches realised the difficulty of competing at the altitude of Mexico City and many of them arrived, and started training, three months before the games started. As a result of this long period of acclimatisation to altitude, the number of red blood cells per unit volume of blood in the athlete's body increased. This is one of the effects of living and working at altitude.
Explain the advantage, to the athletes, of the increased number of red blood cells per unit volume of blood. [2]

....

....

....

8

END OF PAPER